ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Hull Technician (Hull Tech) (00124)
Marine Engineer (Mar Eng) (00367)
Marine Engineering Systems Operator (MESO) (00225)

Gas Free Certification – AFPY

MITE Course IDs: 120998

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commandant of the Curriculum Control Authority (CCA), Canadian Forces Naval Engineering School (CFNES). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J “Course History”.

Revision	Approval Date	Authority
1.0	28 Sept 16	CNPTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated 09 Feb 11;

c. A-PD-055-002/PP-002 Canadian Forces Manual of Military Occupational Structure, Vol 2, Part 2, Occupational Specification for the Hull Technician Occupation, MOC 321 dated 30 Jan 98;

d. A-PD-055-002/PP-002 Marine Engineering Systems Operator (MESO), MOS 00225, Occupation Specification dated 30 Jan 98;

e. A-PD-055-02/PP-002 Job Based Specification Marine Engineer dated 01 Feb 12;

f. A-PD-055-003/PQ-001 Gas Free Certification dated 30 Nov 94; and

g. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the Training Development Center (TDC), who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS A	Gas Free Certification	120998	AFPY
NFS P	Gas Free Certification	120998	AFPY

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for Gas Free Certification. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, para 4.5. For this course, a period of instruction is 50 minutes in duration.

	Periods
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	39.0
Enabling Checks (ECs)	5.5
Performance Checks (PCs)	9.5
Sub-total	54 8 **trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	1.0
Training Administrative Time (TAT)	1.0
Training Travel Time (TTT)	1.0
Physical Fitness (PT)	0
Sub-total	3.0 0 **trg days**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) **8 (55 min)periods** * **Partial days must be rounded up**	57.0 8 **trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Instructor	MS	Gas Free Certified	1	PO 001/002
Instructor	Civilian	FMF Staff- Gas Free FMF Level 3 Certified	1	PO 002

Critical Training Resources	No of Days
References	8
Test gas	8
Gas detector tubes	8
Classroom	8
Colorimetric gas detectors	8
Writing materials	8
One desk and chair for each candidate and instructor	8
White board and markers	8
Podium	8
DWAN Computer/projector	8
Electronic gas detectors	8
Calibration kits	8
Student handouts	8

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS.

a. Maximum – 12;

b. Minimum – 4; and

c. Optimum – 8.

Note: When determining Max, Min and Capacity, the QSPWB is to consult with the TE(S) regarding number of students per course which is based on resource limitations (money, time, instructors to teach and supervise), equipment limitations (boats, tools, workstations), classroom limitations (desks and chairs), safety concerns

(RSO, and Range coaches) and instructional techniques (syndicate size). Operational considerations can be captured if required under Optimum Capacity (4-man attack team, bridge team, RAS team).

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NPTG is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the Training Establishment conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. 00225 MESO/00367 Mar Eng, MS Cert B or above, employed on Naval Reserve Vessels where Hull Techs are not borne;

b. 00367 Mar Eng, MS or above, employed in HMCS Submarines;

c. 00367 Mar Eng, MS or above, employed in units where Hull Techs are not borne;

d. 00124 Hull Technician, QL4 or above;

8. Qualification. Successful completion of this course constitutes the primary means of attaining Gas Free Certification (AFPY). Additionally, the following MITE Course ID will be granted; Gas Free Certification - MITE Course 120998.

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 - Course Reports and Certification.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NPTORD 4500-27 – PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. PO 002 of Gas Free Certification must be completed under the direction of a certified safety section representative of an FMF organization.

14. Additional qualifications exist at FMF and are available for personnel posted as an FMF Safety Inspector. This FMF qualification (currently known as Gas Free Certification Level III) is trained at FMF, and is not detailed nor captured within this QSP as it is outside of the scope.

15. The final performance check (PC 002) in the granting of Gas Free Certification is a Qualification Review Board IAW SEMS.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

PO 001

1. Performance. Prepare Environment ‘Safe to Work’ (QSPWB T0001)

2. Conditions

   a. Given:

      (1) Atmospheric detection/measurement devices;
      (2) Assistance;
      (3) Safety equipment;
      (4) Permits; and
      (5) Specific compartments/spaces to be tested.

   b. Denied: Nil.

   c. Environmental: At sea or alongside in all weather conditions.

3. Standard. The Gas Free Certified personnel shall prepare environment ‘safe to work’ IAW refs, by completing the following:

   a. (QSPWB T0003) Conduct risk assessment by completing the following:

      (1) (QSPWB T0010) Identify specific space hazards; and
      (2) (QSPWB T0011) Coordinate gas free with other ship activities;

   b. (QSPWB T0004) Mitigate identified hazards;

   c. (QSPWB T0005) Establish communication plan; and

   d. (QSPWB T0006) Perform pre-operational equipment checks.

4. References. A1, A2, A3, A5, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16, A17, A19, A20, A21, A22, A23, A24, A25, A26, C1, C2, C3, C4, C5, C6, C7.

5. Task Numbers. QSPWB T0001, QSPWB T0003, QSPWB T0004, QSPWB T0005, QSPWB T 0006, QSPWB T0010, QSPWB T0011.

6. Training Limitations. Nil

PO 002

1. Performance. Certify environment 'safe to work' (QSPWB T0002)

2. Conditions

   a. Given:

      (1) Atmospheric detection/measurement devices;
      (2) Assistance;
      (3) Safety equipment;
      (4) Permits; and
      (5) Specific compartments/spaces to be tested.

   b. Denied: Nil.

   c. Environmental: At sea or alongside in all weather conditions.

3. Standard. The Gas Free Certified personnel shall certify environment safe to work IAW refs, by completing the following:

   a. (QSPWB T0007) Operate detection equipment, by completing the following:

      (1) (QSPWB T0012) conduct air sample; and
      (2) (QSPWB T0013) interpret monitor readings;

   b. (QSPWB T0008) Complete permits IAW SEMS; and

   c. (QSPWB T0009) Complete post certification activities.

4. References. A1, A3, A9, A10, A13, A14, A15, A19, A23, A24, C1, C2, C3, C4, C5, C6.

5. Task Numbers. QSPWB T0002, QSPWB T0007, QSPWB T0008, QSPWB T0009, QSPWB T0012, QSPWB T0013.

6. Training Limitations. An appropriate vessel alongside would be required in order to adequately train this performance objective.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL TRAINING DEVELOPMENT CENTER (TDC)

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of (PRACTICAL or WRITTEN) examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation

5. Standards provide quality control by designing, administering and scoring all Performance Checks. Any PC not administered or invigilated by standards staff will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional and standards staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and TE CSOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the TRB or PRB (if required); and

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment CSOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment CSOs or SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, 4500-51 - Course Reporting and Certificates.

12. Trainee will be graded pass or fail. IAW NPTORD 4500-51 - Course Reporting and Certificates, students will not be ranked.

13. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and / or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL TRAINING DEVELOPMENT CENTER (TDC)

PO 001 Prepare Environment 'Safe to Work' (QSPWB T0001)

EO 001.01 Conduct risk assessment

1. Performance. Conduct risk assessment (QSPWB T0003)

2. Conditions

   a. Given:

      (1) PPE;
      (2) Ship's staff;
      (3) Assistance;
      (4) Atmospheric detection/measurement devices;
      (5) Permits;
      (6) Supervision; and
      (7) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall conduct risk assessment, IAW refs, by completing the following:

   a. (QSPWB T0010) Identify specific space hazards, to include:

      (1) Atmospheric analysis;
      (2) Content identification;
      (3) Physical structure;
      (4) Equipment selection;
      (5) External environmental considerations; and
      (6) Measuring for safe atmosphere.

   b. (QSPWB T0011) Coordinate gas free with other ship activities, to include explanation of permits IAW SEMS, to include:

      (1) S7A – Safe to Enter/Conduct Hot Work in a Confined Space Permit; and
      (2) S7B – Safe to Conduct Hot Work Permit.

4. Time Required

   a. EO - 4 x 50 min period;

   b. EC – 0 x 50 min period (take home written assignment); and

5. Instructional Methods: Interactive lecture.

6. Training Aids

a. DWAN computer/projector

b. PHD 6 CDs BIO TREK II video (ref C6)

7. Test Details

a. EO 001.01 will be tested in EC 001.01 and PC 001;

b. EC 001.01 shall consist of a written take-home test based upon teaching points from para 10. Minimum pass mark is 80%.

c. PC 001 shall consist of:

(1) Part A- written test based upon teaching points from para 10 in EO 001.01- EO 001.04. Minimum pass mark of 80%; and

(2) Part B- practical test based on the application of teaching points from para 10 in EO 001.01- EO 001.04 with a pass/fail IAW PC 001 Part IV Performance Checklist.

8. References. A1, A8, A9, A10, A11, A12, A13, A16, A17, A20, A21, A22, A24, C3, C4, C6, C7.

9. Remarks. All requirements for WHMIS training as described in EO 001.01 and EO 001.02 must be met prior to exposure to any controlled substances.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Identify specific space hazards, to include: • Atmospheric analysis ➢ Methane; ➢ Paints and thinners; ➢ Acetylene; ➢ Chlorine; ➢ PCBs; ➢ Hydrogen sulfide; and ➢ Hydrogen chloride	Interactive Lecture	20	A1, A22, A24
TP2	Identify specific space hazards, to include: • Content identification ➢ Distillate; ➢ JP5; ➢ Gasoline; ➢ Asbestos; ➢ Engulfment; ➢ Entrapment; and ➢ Bridging	Interactive Lecture	20	A1, A12, A22

TP3	Identify specific space hazards, to include: • Physical structure ➢ Layout; ➢ Entry/exit; ➢ Pipes/lights; ➢ Magazines; and ➢ Fall arrest staging	Interactive Lecture	20	A1, A8, A20, A24
TP4	Identify specific space hazards, to include: • Equipment selection ➢ Intrinsically safe	Interactive Lecture	20	A1
TP5	Identify specific space hazards, to include: • External environmental considerations ➢ Ice ➢ Condensation	Interactive Lecture	20	A1, A20
TP6	Identify specific space hazards, to include: • Measuring for safe atmosphere ➢ Ignition; ➢ Flash point; ➢ LEL/UEL; ➢ TLV; ➢ TWA; ➢ Stel; ➢ Carbon; ➢ IDLH (immediately dangerous to life and health)	Interactive Lecture	50	A1, A17, A21, A22, C3, C4
TP7	Coordinate gas free with other ship activities to include: • Explanation of permits IAW SEMS ➢ Hot work; ➢ Confined space; ➢ Man aloft; ➢ Lock out tag out; and ➢ Radiation hazard	Interactive Lecture	50	A24

Enabling check - EC 001.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. One EC test per student.

Arrangements and Procedures

2. EC 001.01 will be a take-home written test assessment based on content from EO 001.01.

3. Pass / Fail or Pass mark will be 80%.

4. Time required: n/a - take home written test assignment.

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a take-home written test assessment based on content from EO 001.01.

2. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet (this improves feedback given in preparation of the PC).

3. Read each question carefully, and review your response before submitting the test.

4. References will be provided.

5. Pass / Fail or Pass mark is 80%.

6. You will have until the beginning of the next class to complete this assessment.

7. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. This take-home test shall test the ability to understand the materials and not just remember or recall/extract data from references and/or class notes. The questions should require the student to construct meaning from instructional messages, including oral, written and graphic communication. The student will be required to paraphrase, classify, summarize and infer in order to reach the correct answers by using the materials presented in EO 001.01. Multiple versions of the test, for delivery during each serial are recommended.

PO 001 Prepare Environment 'Safe to Work' (QSPWB T0001)

EO 001.02 Mitigate identified hazards

1. Performance. Mitigate identified hazards (QSPWB T0004)

2. Conditions

   a. Given:

      (1) Atmospheric detection/measurement devices;
      (2) Permits;
      (3) Assistance;
      (4) Safety equipment;
      (5) Supervision; and
      (6) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall mitigate identified hazards, IAW refs, by completing the following:

   a. Mitigate for confined space, to include:

      (1) Gather initial PPE;
      (2) Space configuration; and
      (3) Post restricted area signs.

   b. Mitigate for hot work; to include:

      (1) Identify hot work restrictions; and
      (2) Compartment preparation.

4. Time Required

   a. EO - 4 x 50 min period;

   b. EC – 0 x 50 min period (take home written assignment); and

5. Instructional Methods: Interactive lecture

6. Training Aids

   a. DWAN Computer/projector

   b. PHD 6 CDs BIO TREK II video (ref C6)

7. Test Details

a. EO 001.02 will be tested in EC 001.02 and PC 001;

b. EC 001.02 shall consist of a written take-home test based upon teaching points from para 10. Minimum pass mark is 80%.

c. PC 001 shall consist of:

(1) Part A- written test based upon teaching points from para 10 in EO 001.01- EO 001.04. Minimum pass mark of 80%; and

(2) Part B- practical test based on the application of teaching points from para 10 in EO 001.01- EO 001.04 with a pass/fail IAW PC 001 Part IV Performance Checklist.

8. References. A1, A2, A3, A5, A9, A10, A13, A14, A15, A19, A24, C6.

9. Remarks. All requirements for WHMIS training as described in EO 001.01 and EO 001.02 must be met prior to exposure to any controlled substances.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Mitigate for confined space, to include: • Gather initial PPE ➢ Harness inspection IAW Fall Arrest training	Interactive Lecture	40	A1, A24
TP2	Mitigate for confined space, to include: • Space configuration ➢ Set up ventilation ➢ Lock out tag out ➢ Check space is empty as possible	Interactive Lecture	40	A1, A24
TP3	Mitigate for confined space, to include: • Post restricted area signs	Interactive Lecture	20	A1
TP4	Mitigate for hot work, to include: • Identify hot work restrictions ➢ Ammunition ➢ Man aloft considerations	Interactive Lecture	20	A1, A24

TP5	Mitigate for hot work, to include: • Compartment preparation ➢ Insulation removal ➢ Fire cloth ➢ Inerting ➢ Fire extinguishers ➢ Flammable cover ➢ Adjacent compartment rounds ➢ Intrinsically safe tools ➢ Isolate fitted systems ➢ Set up ventilation "point" ➢ Lock out tag out ➢ Lead test insulation	Interactive Lecture	80	A1, A24

Enabling check - EC 001.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. One EC test per student.

Arrangements and Procedures

2. EC 001.02 will be a take-home written test assessment based on content from EO 001.02.

3. Pass / Fail or Pass mark will be 80%.

4. Time required: n/a – take home written test assignment.

5. References will be required

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a take-home written test assessment based on content from EO 001.02.

2. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet (this improves feedback given in preparation of the PC).

3. Read each question carefully, and all responses fully, before answering the question.

4. References will be provided.

5. Pass / Fail or Pass mark is 80%.

6. You will have until the beginning of the next class to complete this assessment.

7. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. This take-home test shall test the ability to understand the materials and not just remember or recall/extract data from references and/or class notes. The questions should require the student to construct meaning from instructional messages, including oral, written and graphic communication. The student will be required to paraphrase, classify, summarize and infer in order to reach the correct answers by using the materials presented in EO 001.02. Multiple versions of the test, for delivery during each serial are recommended.

PO 001 Prepare Environment 'Safe to Work' (QSPWB T0001)

EO 001.03 Establish communication plan

1. Performance. Establish communication plan (QSPWB T0005)

2. Conditions

   a. Given:

      (1) Ship's staff;
      (2) Atmospheric detection/measurement devices;
      (3) Permits;
      (4) Supervision;
      (5) Assistance;
      (6) PPE; and
      (7) Safety equipment.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall establish communication plan, IAW refs, by completing the following:

   a. Brief CoC and OPI;

   b. Brief the assigned team members; and

   c. Formulate rescue plan.

4. Time Required

   a. EO - 2 x 50 min period;

   b. EC – 0.5 x 50 min period; and

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration – performance.

6. Training Aids

   a. DWAN computer/projector

   b. Permits

7. Test Details

a. EO 001.03 will be tested in EC 001.03 (in conjunction with EO. 001.04) and PC 001;

b. EC 001.03 shall consist of a practical test based upon teaching points from para 10. Pass/Fail IAW Part IV Enabling Checklist.

c. PC 001 shall consist of:

(1) Part A- written test based upon teaching points from para 10 in EO 001.01- EO 001.04. Minimum pass mark of 80%; and

(2) Part B- practical test based on the application of teaching points from para 10 in EO 001.01- EO 001.04 with a pass/fail IAW PC 001 Part IV Performance Checklist.

8. References. A1, A8, A24

9. Remarks. All requirements for WHMIS training as described in EO 001.01 and EO 001.02 must be met prior to exposure to any controlled substances.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Establish communication plan by: • Brief CoC and OPI ➢ Make pipes	Interactive Lecture/Demo	20	A1, A24
TP2	Establish communication plan by: • Brief the assigned team members ➢ Communications check	Interactive Lecture/Demo	20	A1, A24
TP3	Establish communication plan by: • Formulate rescue plan ➢ Situational assessment • Responsibilities of: ➢ CoC/HQ1 ➢ RAT ➢ OSC	Interactive Lecture/Demo	60	A1, A8, A24

PO 001 Prepare Environment 'Safe to Work' (QSPWB T0001)

EO 001.04 Perform pre-operational equipment checks

1. Performance. Perform pre-operational equipment checks (QSPWB T0006)

2. Conditions

   a. Given:

      (1) Atmospheric detection/measurement devices;
      (2) Supervision;
      (3) Permits;
      (4) Ship's staff;
      (5) Assistance; and
      (6) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall perform pre-operational equipment checks, IAW refs, by completing the following:

   a. Conduct pre-operational checks on the safety equipment, to include:

      (1) PPE;
      (2) Communications;
      (3) Rescue equipment;
      (4) Staging requirements; and
      (5) Ventilation fans.

   b. Conduct pre-operational equipment checks on gas detectors, to include:

      (1) Perform pre-operational checks on gas detector; and
      (2) Perform pre-operational checks on colorimetric detector.

4. Time Required

   a. EO - 8 x 50 min period;

   b. EC – 0.5 x 50 min period; and

   c. PC – 2.5 x 50 min period.

5. Instructional Methods:

   a. Interactive lecture; and

   b. Demonstration – performance.

6. Training Aids

a. PhD6;

b. Drager;

c. Tubes;

d. References;

e. PPE;

f. DWAN computer/projector; and

g. Mock-up.

7. Test Details

a. EO 001.04 will be tested in EC 001.03 and PC 001;

b. EC 001.03 shall consist of a practical test based upon teaching points from para 10. Pass/Fail IAW Part IV Enabling Checklist.

c. PC 001 shall consist of:

(1) Part A- written test based upon teaching points from para 10 in EO 001.01- EO 001.04. Minimum pass mark of 80%; and

(2) Part B- practical test based on the application of teaching points from para 10 in EO 001.01- EO 001.04 with a pass/fail IAW PC 001 Part IV Performance Checklist.

8. References. A1, A7, A8, A15, A17, A23, A24, A25, A26, C1, C2, C3, C4, C5, C6.

9. Remarks. Nil.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Conduct pre-operational checks on the safety equipment, to include: • PPE	Interactive Lecture/Demo	10	A1, A15, A17, A23
TP2	Conduct pre-operational checks on the safety equipment, to include: • Communications	Interactive Lecture/Demo	10	A1, A25, A26
TP3	Conduct pre-operational checks on the safety equipment, to include: • Rescue equipment	Interactive Lecture/Demo	10	A1, A8, A24

TP4	Conduct pre-operational checks on the safety equipment, to include: • Staging requirements	Interactive Lecture/Demo	10	A1, A7, A8, A24
TP5	Conduct pre-operational checks on the safety equipment, to include: • Ventilation fans	Interactive Lecture/Demo	10	A1, A8, A24
TP6	Conduct pre-operational equipment checks on gas detectors, to include: • Pre-operational checks on gas detector, which includes: ➢ Explain function and operations ➢ ID components ➢ Calibrate bump test ➢ Remote sampling ➢ Maintenance	Interactive Lecture/Demo	300	A1, C3, C6
TP7	Conduct pre-operational equipment checks on gas detectors, to include: • Perform pre-operational checks on colorimetric detector, which includes: ➢ List tubes available and how to use each ➢ Explain reference materials	Interactive Lecture/Demo	50	A1, C1, C2, C4, C5

Enabling check - EC 001.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. PhD6, Drager, tubes, references, PPE, and mock-up, and one EC checklist per student.

Arrangements and Procedures

2. EC 001.03 will be a practical assessment based on content from EO 001.03 and EO 001.04.

3. Pass / Fail IAW EC 001.03 Part IV Enabling Checklist.

4. Time required: 0.5 x 50 min period(s). Trainees are assessed concurrently throughout the practical phase time allotted IAW EC 001.03 Part IV Checklist.

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a practical assessment based on content from EO 001.03 and EO 001.04.

2. During this practical assessment, you will be required to perform pre-operational equipment checks. You must pass this test to pass EO 001.03 and EO 001.04. To pass the test, you must perform pre-operational equipment checks in accordance with the EC 001.03 Part IV Enabling Checklist by completing the following:

   a. Coordinate communications;

   b. Conduct pre-operational checks on the safety equipment; and

   c. Conduct pre-operational equipment checks on gas detectors.

3. References will be provided.

4. Pass / Fail IAW checklist.

5. You will have 0.5 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. SCENARIO: Trainee is to be informed that he/she will be doing a gas test on a compartment/tank using manual and automatic instruments. Instruct the trainee that you require an Air Quality and Oxygen Content Test. Trainee will not enter the tank.

2. All steps are critical. Trainee must prove to testing authority that he /she is capable of carrying out a gas test of a compartment/tank without assistance or prompting from instructor. If assistance is given, this is to be noted. The trainee will then be advised of his/her shortcomings.

PART IV – ENABLING CHECK LIST

Check List	*SAT*	*UNSAT*	*Comments*
Turn PHD6 and wait at least five (5) minutes to allow the readings to fully stabilize			
Conduct fresh air/zero calibration			
Conduct gas calibration “bump test”			
Remove and replace the battery pack			
Remove and replace the PID sensor			
Perform “remote sampling” with the sample probe and hose assembly.			
Test pump for leaks			
Purge Drager pump in fresh air			
Insert polytest tube into pump, (verify directional arrow)			
Place polytest tube into compartment and pump required no. of aspirations (located on tube)			
Examine polytest tube (with the aid of info sheet)			
Insert oxygen tube, (verify directional arrow and filter tube position)			
Lower oxygen tubes into comp't and pump required no. of aspirations (1)			
Examine oxygen tube (with the aid of info sheet)			

The trainee must receive 14 SAT to pass this EC.

PERFORMANCE CHECK - PC 001

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom, PhD6, Drager, tubes, references, PPE, mock-up,one PC exam per student, and one PC checklist per student.

Arrangements and Procedures

2. PC 001 will be:

   a. Part A: a written test based on content from PO 001; and

   b. Part B: a practical test based on content from PO 001.

3. Pass / Fail or Pass mark will be:

   a. Part A 80%; and

   b. Part B Pass/Fail IAW PC001 Part B Part IV Performance Checklist.

4. Time required:

   a. Part A- 1.0 x 50 min period; and

   b. Part B- 1.5 x 50 min period

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE (PC 001- Part A)

1. PC 001- Part A will be a written test based on content from PO 001.

2. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet .

5. Work carefully. Do not spend too much time trying to answer questions(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass / Fail or Pass mark will be 80%.

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE (PC 001- Part B)

1. PC 001- Part B will be a practical assessment based on content from PO 001.

2. During this practical assessment, you will be required to prepare environment 'safe to work'. You must pass this test to pass PO 001. To pass the test, you must prepare environment 'safe to work' in accordance with the PC 001- Part B checklist by completing the following:

   a. conduct risk assessment;

   b. mitigate identified hazards;

   c. establish communication plan; and

   d. perform pre-operational equipment checks.

3. References will be provided.

4. Pass / Fail IAW checklist.

5. You will have 1.5 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. SCENARIO: Trainee is to be informed that he/she will be doing a gas test on a compartment/tank using manual and automatic instruments. Instruct the trainee that you require an Air Quality and Oxygen Content Test. Trainee will not enter the tank.

2. All steps are critical. Trainee must prove to testing authority that he /she is capable of carrying out a gas test of a compartment/tank without assistance or prompting from instructor. If assistance is given, this is to be noted. The trainee will then be advised of his/her shortcomings and will then be given the opportunity to do the PC after everyone else is finished. If the trainee fails again, it will be assessed as a failure and Standards is to be informed.

PART IV – PERFORMANCE CHECK LIST

Check List	*Yes*	*No*	*Comments*
Turn PHD6 and wait at least five (5) minutes to allow the readings to fully stabilize			
Conduct fresh air/zero calibration			
Conduct gas calibration “bump test”			
Remove and replace the battery pack			
Remove and replace the PID sensor			
Perform “remote sampling” with the sample probe and hose assembly.			
Test pump for leaks			
Purge Drager pump in fresh air			
Insert polytest tube into pump, (verify directional arrow)			
Place polytest tube into compartment and pump required no. of aspirations (located on tube)			
Examine polytest tube (with the aid of info sheet)			
Insert oxygen tube, (verify directional arrow and filter tube position)			
Lower oxygen tubes into comp't and pump required no. of aspirations (1)			
Examine oxygen tube (with the aid of info sheet)			

The trainee must receive 14 SAT to pass this PC.

PO 002 Certify Environment 'Safe to Work' (QSPWB T0002)

EO 002.01 Operate detection equipment

1. Performance. Operate detection equipment (QSPWB T0007)

2. Conditions

   a. Given:

      (1) Assistance;
      (2) Supervision;
      (3) Safety equipment;
      (4) Specific compartment/space to be tested;
      (5) Atmospheric detection/measurement devices; and
      (6) Ship's staff.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom;
      (2) At sea or alongside in all weather conditions.

3. Standard. The trainee shall operate detection equipment, IAW refs, by completing the following:

   a. Conduct air sample (QSPWB T0012), to include:

      (1) Operate gas detector; and
      (2) Operate colorimetric gas detector.

   b. Interpret monitor readings (QSPWB T0013) by comparison to limits IAW ACGIH TLV/BEI indexes.

4. Time Required

   a. EO - 7 x 50 min period;

   b. EC – 5.0 x 50 min period; and

   c. PC – 7.0 x 50 min period.

5. Instructional Methods:

   a. Demonstration – performance; and

   b. Guided discussion.

6. Training Aids

a. PhD6;

b. Drager;

c. Tubes;

d. References;

e. PPE;

f. DWAN computer/projector; and

g. Mock-up.

7. Test Details. EO 002.01 will be tested in EC 002.01 and tested in PC 002, which is the final Qualification Review Board.

8. References. A1, C1, C2, C2, C3, C4, C5, C6.

9. Remarks. The trainee will also be provided with pre-QRB practice questions as part of group discussions.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Conduct air sample, to include: • Operate generic gas detector IAW manufacturer's instructions, which includes: ➢ Verify calibration and bump test ➢ Set up for remote samples ➢ Take samples	Practical	125	A1, C3, C6
TP2	Conduct air sample, to include: • Operate colorimetric gas detector IAW manufacturer's instructions, which includes: ➢ Select appropriate tube ➢ Follow manufacturer's instructions ➢ Take sample	Practical	125	A1, C1, C2, C4, C5
TP3	Interpret monitor readings by comparison to limits IAW ACGIH TLV/BEI indexes	Practical	100	A1, C1, C2, C3, C4, C5, C6,

PO 002 Certify Environment 'Safe to Work' (QSPWB T0002)

EO 002.02 Complete permit IAW SEMS (QSPWB T0008)

1. Performance. Complete permit IAW SEMS (QSPWB T0008)

2. Conditions

   a. Given:

      (1) Specific compartment/space to be tested;
      (2) Supervision;
      (3) Ship's staff;
      (4) Assistance;
      (5) Permits;
      (6) Atmospheric detection/measurement devices; and
      (7) Safety equipment..

   b. Denied: Nil.

   c. Environmental: At sea or alongside in all weather conditions.

3. Standard. The trainee shall complete permits IAW ref A23, by completing the following:

   a. Complete S7A: Safe to Enter/Conduct Hot Work in a Confined Space Permit;

   b. Maintain permits; and

   c. Complete S7B: Safe to Conduct Hot Work Permit.

4. Time Required

   a. EO - 7 x 50 min period;

   b. EC – 5.0 x 50 min period; and

   c. PC – 7.0 x 50 min period.

5. Instructional Methods:

   a. Demonstration – performance; and

   b. Guided discussion.

6. Training Aids

   a. PhD6;

   b. Drager;

c. Tubes;

d. References;

e. PPE;

f. DWAN computer/projector; and

h. Mock-up.

7. Test Details. EO 002.02 will be tested in EC 002.01 and in PC 002, which is the final Qualification Review Board.

8. References. A1, A3, A9, A10, A13, A14, A15, A19, A23, A24.

9. Remarks. The trainee will also be provided with pre-QRB practice questions as part of group discussions.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Complete S7A: Safe to Enter/Conduct Hot Work in a Confined Space Permit	Practical	125	A1, A3, A9, A10, A13, A14, A15, A19, A 23, A24
TP2	Maintain permits	Practical	125	A1, A3, A9, A10, A13, A14, A15, A19, A 23, A24
TP3	Complete S7B: Safe to Conduct Hot Work Permit	Practical	100	A1, A3, A9, A10, A13, A14, A15, A19, A23, A24

PO 002 Certify Environment 'Safe to Work' (QSPWB T0002)

EO 002.03 Complete post certification activities (QSPWB T0009)

1. Performance. Complete post certification activities (QSPWB T0009)

2. Conditions

   a. Given:

      (1) Atmospheric detection/measurement devices;
      (2) Confined space/hot work chit binders;
      (3) Ship's staff;
      (4) Safety equipment;
      (5) Specific compartment/space to be tested;
      (6) Supervision;
      (7) Permits; and
      (8) Assistance.

   b. Denied: Nil.

   c. Environmental: At sea or alongside in all weather conditions.

3. Standard. The trainee shall complete post certification activities, IAW refs, by completing the following:

   a. Log permits IAW SEMS;

   b. Monitor data logs; and

   c. Maintain UER personnel records through Monitor Mass.

4. Time Required

   a. EO - 7 x 50 min period;

   b. EC – 5.0 x 50 min period; and

   c. PC – 7.0 x 50 min period.

5. Instructional Methods:

   a. Demonstration – performance; and

   b. Guided discussion.

6. Training Aids

   a. PhD6;

   b. Drager;

c. Tubes;

d. References;

e. PPE;

f. DWAN computer/projector; and

h. Mock-up.

7. Test Details. EO 002.03 will be tested in EC 002.01 and PC 002, which is the final Qualification Review Board.

8. References. A1, A3, A9, A10, A13, A14, A15, A19, A23, A24.

9. Remarks. The trainee will also be provided with pre-QRB practice questions as part of group discussions.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Complete post certification activities, to include: • Log permits IAW SEMS • Monitor data logs • Maintain UER personnel records through Monitor Mass	Practical	350	A1, A3, A9, A10, A13, A14, A15, A19, A23, A24

Enabling check - EC 002.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. PhD6, Drager, tubes, references, PPE, mock-up, and one EC checklist per student.

Arrangements and Procedures

2. EC 002.01 will be a practical assessment based on content from EO 002.01 to EO 002.03.

3. Pass / Fail IAW EC 002.01 Part IV Enabling Checklist.

4. Time required: 5.0 x 50 min period(s). Trainees are assessed concurrently throughout the practical phase time allotted IAW EC 002.01 Part IV Checklist.

5. References will be required:

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a practical assessment based on content from EO 002.01 to EO 002.03.

2. During this practical assessment, you will be required to certify environment 'safe to work'. You must pass this test to pass EO 002.01 to EO 002.03. To pass the test, you must certify environment 'safe to work' in accordance with the EC 002.01 Part IV Enabling Checklist by completing the following:

   a. Operate detection equipment;

   b. Complete permit IAW SEMs; and

   c. Complete post certification activities.

3. References will be provided.

4. Pass / Fail IAW checklist.

5. You will have 5 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. SCENARIO: Trainee is to be informed that he/she will be doing gas tests on compartments/tanks using manual and automatic instruments. Instruct the trainee that you require an Air Quality and Oxygen Content Test.

2. All steps are critical. Trainee must prove to testing authority that he /she is capable of carrying out a gas test of a compartment/tank without assistance or prompting from instructor. If assistance is given, this is to be noted.

PART IV – ENABLING CHECK LIST

Check List	*SAT*	*UNSAT*	*Comments*
CONFINED SPACES- Trainee to certify a confined space and provide a risk assessment on a black or grey water tank			
CONFINED SPACES- Fuel or Oil tank			
CONFINED SPACES- Fresh water tank or ballast tank or void or a cofferdam			
HOT WORK CHITS- Trainee to carry out a hot work certification and risk assessment on a machinery space			
HOT WORK CHITS- Mess decks or living spaces			
HOT WORK CHITS- Upper deck			
HOT WORK CHITS- Area adjacent to a magazine			

The trainee must receive 7 SAT to pass this EC.

PERFORMANCE CHECK - PC 002

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Board Room, QRB Board personnel as per ROD para 34, one PC Checklist per student and QRB board member and one Training Completion Certificate (Annex E) per student.

Arrangements and Procedures

2. PC 002 will be an oral test based on content from PO 002 in the form of a Qualification Review Board. QRB composition will be the School's Hull Training Chief (or their delegate), Section Head (acting as board chair), and a qualified Level III Safety and Environment Inspector, as appointed by the FMF Safety Environment Manager.

3. Training Completion Certification (Annex E) completed upon successful challenge of the AFPY QRB IAW SEMS.

4. Pass / Fail

5. Time required: 7.0 x 50 min period.

6. References will not be required.

PART II – ORAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be an oral assessment based on content from PO 002.

2. During this oral assessment, you will be required to verbally demonstrate your knowledge to certify an environment 'safe to work'. You must pass this test to pass PO 002. To pass the test, you must demonstrate your knowledge to certify an environment 'safe to work' in accordance with PC 002 checklist.

3. References will not be provided.

4. Pass / Fail IAW PC 002 Checklist.

5. You will have 0.5 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Details of what to review and what will be questioned during the AFPY QRB are obtained by reviewing the Performance Objectives in Chapter 2 and the Enabling Objectives in Chapter 4 of this QSP, as well as PC 002 Part IV – Performance Check List.

2. A trainee must obtain a pass in PC 001 and successfully challenge the AFPY QRB in order to be awarded the Gas Free Certification Qualification.

3. QRB composition will be the School's Hull Training Chief (or their delegate), Section Head (acting as board chair), and a qualified Level III Safety and Environment Inspector, as appointed by the FMF Safety Environment Manager. The trainee shall be required to sit and successfully pass a qualification board before being awarded the Gas Free Certification Qualification. Board Chairperson and members must be aware of the content of the training received and are cautioned regarding going beyond the intended scope of the qualification program.

4. Training Completion Certification Form is found in Annex E of this QSP.

PART IV – PERFORMANCE CHECK LIST

Check List	*Yes*	*No*	*Comments*
Certify SAFE TO ENTER- Fuel Tanks			
Certify SAFE TO ENTER- Quiet Medium Tank			
Certify SAFE TO ENTER- Salt Water Ballast Tanks			
Certify SAFE TO ENTER- Grey Water Tanks			
Certify SAFE TO ENTER- Black Water			
Certify SAFE TO ENTER- Chain Locker			
Certify SAFE TO ENTER- C5 Dome			
Certify for HOT WORK- Flight Deck and Hangar			
Certify for HOT WORK- Mess Deck			
Certify for HOT WORK- Foc'sle			
Certify for HOT WORK- Upper Decks			
Certify for HOT WORK- Bridge Wings			
Certify for HOT WORK- Hangar Top			
Certify for HOT WORK- Bridge Top			

The trainee must receive 14 SAT to pass this PC.

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	DND General Safety Standards
A2	Maritime Command Precautions and Safety Practices for Welding and Burning
A3	Naval Engineering Manual
A4	Not allocated
A5	Safety Precautions and Accident Prevention Instructions Explosives Safety Manual Vol III Ships
A6	Victoria Submarine Technical Instructions VSTI 208
A7	Canada Labor Code
A8	Damage Control Manual, Vol I and 4
A9	Ships' Standing Orders
A10	NAVORD 1002-0
A11	Maintenance Painting Specifications for HMC Ships
A12	Canadian Occupational Safety and Health Regulations, PartXI
A13	Marine Occupational Safety and Health Regulations, Part XI
A14	Victoria Class Submarine Standing Orders AL9
A15	GENERAL SAFETY TRAINING
A16	GENERAL SAFETY PROGRAM HAZARDOUS MATERIALS SAFETY AND MANAGEMENT MANUAL
A17	DND Respiratory Protection Program
A18	Not allocated
A19	Treasury Board Manual Occupational Safety and Health Chapter 2-7, Confined Space Entry
A20	DND/CF Guide to Hazard Analysis
A21	DND/CF Guide to Hazard Identification Health and Safety Risk Assessments
A22	Shipboard Hazardous Material Portfolio
A23	Formation Safety and Environmental Management System Manual
A24	Ship Class Safety Environmental System Manual
A25	Comsec Material Control Policies Standard and Procedures

A26	STU iii Student Handbook
B1	Not allocated
C1	Drager Manufacturer's Detector Tube Handbook
C2	Manufacturer's Instructions (Detector Tube Packets)
C3	PhD 6 Manufacturer's Instructions and Application Notes
C4	Drager/Auer Manufacturer's Instructions
C5	Drager/Auer Threshold Limit Values Manual
C6	PhD 6 CDs BIO TREK II, MANUAL, PhD 6 TRAINING
C7	American Conference of Government and Industrial Hygenists Guide to Threshold Limit Values and Biological Exposure Indecies

ANNEX B – TASK LIST

Task No	Task Statement	Trg / No Trg	PO
QSPWBT0001	Prepare environment ‘safe to work’	Trg	001
QSPWBT0002	Certify environment ‘safe to work’	Trg	002
QSPWBT0003	Conduct risk assessment	Trg	001
QSPWBT0004	Mitigate identified hazards	Trg	001
QSPWBT0005	Establish communication plan	Trg	001
QSPWBT0006	Perform pre-operational equipment checks	Trg	001
QSPWBT0007	Operate detection equipment	Trg	002
QSPWBT0008	Complete permit IAW SEMS	Trg	002
QSPWBT0009	Complete post certification activities	Trg	002
QSPWBT0010	Identify specific space hazards	Trg	001
QSPWBT0011	Coordinate gas free with other ship activities	Trg	001
QSPWBT0012	Conduct air sample	Trg	002
QSPWBT0013	Interpret monitor readings	Trg	002

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates:

13 - 24 July 2015

Location:

CFNES, Bldg S-9, LM2500 Classroom

Board Members:

Surname	Initials	Rank	MOSID	Unit
Yakimovich	K.A.	Lt (N)	00345	NPTG HQ
Phillis	P.K.	Lt (N)	00345	MARLANT FSE
Rivalin	G.C.	Capt	00211	CFNES
Townsend	C.L.	SLt	00211	CFNES
Faludi	C.M.	SLt	00211	CFNES
Hynes	W.G.	SLt	00345	CFNES
Simmons	R.A.	SLt	00345	CFNOS
Burton	N.R.	CPO2	00366	NPTG HQ
Wood	G.K.	PO1	00367	CFNOS
Ross	J.W.	PO1	00124	CFNES
Forsyth	R.D.	PO1	00124	CFNES
Holt	D.T.	PO2	00124	CFFSE
Caldwell	J.M.	PO2	00124	CFNES
MacLean	J.T.	Civilian	n/a	MARPAC FSE
Kelly	W.M.	Civilian	n/a	MARLANT FSE
Caird	R.O.	Civilian	n/a	FMF Cape Breton
Osmond	R.G.	Civilian	n/a	FMF Cape Scott

Board Objective:

1. To review, update and amend course content to:

    a. ensure all levels of qualification requirements are captured within the document;

    b. ensure the QSP takes into consideration civilian rules and still falls within the RCN scope;

c. ensure the gas free standard meets both Canadian and Maritime Occupational Health and Safety Regulations;

d. identify and delete obsolete training elements;

e. identify and add new training elements;

f. establish measurable standards;

g. assess critical outstanding QSP amendments and integrate necessary changes into QSP;

h. finalize composition of QSP; and

i. review and amend QSP formatting IAW CFITES.

Specification Task – No Train Decisions:

Task No	Task Statement	Substantiation
T0001	Certify compartment "Gas Free"	Current SS includes only one vague task, which does not accurately reflect on-job performance. The Board therefore generated 13 new tasks. NPTG Reps liaised with DNavP&T Occupation Managers and NPTG for approval of this new list prior to the board moving onto the Design Phase.

Discussions / Decisions:

Day #1 - 13 Jul 2015

1. QSPWB commences @ 0830 and briefs were given by:

a. SLt Hynes on the QSPWB process;

b. SLt Townsend on the Analysis Phase of CFITES; and

c. Capt Rivalin on Integrated Systems Approach to Training (ISAT) Training Document Management System.

2. Members were given time to review/discuss reference materials.

3. The Board discussed the difference between the current three levels of gas free certification and differences between military and civilian qualifications and jobs performed depending on circumstance, such as at sea or alongside.

4. CPO2 Burton, along with CPO2 Overs will liaise with DNavP&T Occupation Managers IOT get support for the decision to make this Gas Free Certification QSP to include a 3 day residential phase followed by a 5 day practical phase, with the practical phase being completed under the guidance of a FMF Safety Inspector. Upon

completion of the eight days, the member will be granted the NQual (AFPY). John MacLean also discussed this decision with RCN General Safety Officer, Jim Dzairski and he agreed with this decision. The board felt that the current level 1 training in and of itself does not prepare members for on-job performance. It was felt this QSP does require a mention that FMF will still provide additional levels of recognition, however FMF recognitions fall outside the scope of this QSP and shall not be detailed.

5. Personnel to receive this training include the following:

a. 00124 Hull Technician, QL5A or above;

b. 00367 Mar Eng, MS or above, employed in HMCS Submarines;

c. 00367 Mar Eng, MS or above, employed in units where H Techs are not borne;

d. 00225 MESO/00367 Mar Eng, MS Cert B or above, employed on Naval Reserve Vessels where H Techs are not borne;

e. Personnel employed at an FMF Safety Section; and/or

f. DND civilian personnel in possession of a 4th Class Engineering Certificate.

6. SLt Townsend led the board through Task Analysis IOT generate a list of tasks requiring training in order for a member to become gas free certified.

7. CFNES Commandant, Cdr Lebel provided a brief on the NPTG Engine, and the need to use the resources we have IOT not increase days of training.

Day #2 - 14 Jul 2015

8. The Board continued with Task Analysis.

9. Unanimous agreement received from NPTG TMs that the current 5 day OJT will become a practical phase of the course, as it was not appropriately named. OJT is when an employee is training at the place of work while he or she is doing the actual job and this is not what happens during this practical phase and thus will not be referred to as OJT.

10. The Board discussed whether "re-certification" should occur every ninety days, as per MARCORD 66-01 or increase to 180 or 365 days. Idea of having a chit to serve same purpose is also discarded. The length of time that re-certification should cover is put in parking lot for later discussion.

Day #3 - 15 Jul 2015

11. The Board discussed the scope of the course and the fact that it should not give authorization for confined space entry.

12. The Board discussed the need for a new MITE code in order to avoid any confusion with members who currently hold Level 1 and/or Level 2 training. The current MITE code will be archived. The same national qualification will be able to be linked to the new MITE code (AFPY).

13. The Board re-visited the discussion about additional level of qualification available at FMF and the need to consider adding an Experience Qualification (EQual) for military members posted to FMF who meet the FMF requirements for the additional level of qualification available at FMF. No decision was reached on this topic.

14. Chapters 1 & 2 drafted and approval obtained from NPTG via Training Managers, Lt(N) Yakimovich and CPO2 Burton for the 13 tasks identified, which include:

a. prepare environment 'safe to work' (QSPWBT0001);

b. certify environment 'safe to work' (QSPWBT0002);

c. conduct risk assessment (QSPWBT0003);

d. mitigate identified hazards (QSPWBT0004);

e. establish communication plan (QSPWBT0005);

f. perform pre-operational equipment checks (QSPWBT0006);

g. operate detection equipment (QSPWBT0007);

h. complete permit IAW SEMS (QSPWBT0008);

i. complete post-certification activities (QSPWBT0009);

j. identify specific space hazards (QSPWBT0010);

k. coordinate gas free with other ship activities (QSPWBT0011);

l. conduct air sample (QSPWBT0012); and

m. interpret monitor readings (QSPWBT0013).

15. The Board completed work on Chapters 1 & 2, which included:

a. the decision to change the training limitations of "operational ships" to "vessel" to encompass other resources that may have to be utilized, such as de-commissioned ships, tanks, etc.; and

b. the decision to change the course capacity to a minimum of 4, a maximum of 12 and an optimum of 8 students with a note stating the course may be required to run below minimum due to operational requirements of the fleet.

16. The Board re-visited the discussion of whether to endorse certification every 90 or 180 days. Provincial standard is every two years. A decision was not reached.

17. The Board discussed whether a pre-requisite of Confined Space Entry needs to be taken before the Gas Free Certification course. It was decided that the previous knowledge students will bring to the course suffices on the subject of confined space entry, and thus the pre-requisite is not required.

Day #4 - 16 Jul 2015

18. The decision is made that recertification process will be followed IAW the new NAVORD 1002-0. A discussion about what will be recommended for the NAVORD takes place with no decision made at this time.

19. SLt Townsend delivers a brief on the Design Phase of CFITES.

20. References were verified and added as required.

21. Instructional Analysis (IA) begins and is completed for PO002. IA begins for PO001.

Day #5 - 17 Jul 2015

22. PO2 Caldwell makes the recommendation for a deviation from the normal DND configuration of the PhD 6 to have a CO H sensor for use on submarines and will action this item.

23. The Board recommends the Qualification Review Board (QRB) composition will be the School's Hull Training Chief, Section Head (acting as board chair), a Qualified Level III Safety and Environment Inspector as appointed by the FMF Safety Environment Manager.

24. The Board is recommending to the RCN GSO, Jim Dzairski that the "endorsement" time be amended to 180 days and tracked through Monitor Mass to ensure proficiency. This would allow the unit training chief to track this qualification on a monthly basis. Sea training has ensured that readiness opportunities will be provided inspectors to maintain proficiency during CRR.

25. A conversation occurred that affirmed this board's intention is by no means attempting to change the policy in place that Gas Free Certifications occurring alongside is performed by FMF Safety Inspectors.

26. The board is recommending the creation of a DLN module to cover the theory portion in advance of support of endorsement of the qualification. This course, once developed, will be completed if endorsement is not obtained within 180 days. Once the DLN module is completed, the member would then do a practical assessment with FMF (as is the current policy for re-endorsement). Should a technician pass a two year point without endorsing his qualification, they must complete the DLN module, complete a written exam with an instructor and then obtain a practical assessment endorsement by the Section Head, under recommendation by FMF. Should the qualification expire after 5 years without endorsement the entire course must be completed.

Day #6 - 20 Jul 2015

27. Week #2 commences with the following board members present:

a. SLt Hynes (Chairperson);

b. SLt Townsend (TDO);

c. SLt Faludi (Secretary);

d. Capt Rivalin (TDO OJT Supervisor);

e. PO2 Caldwell (SME); and

f. PO1 Forsyth (SME).

28. The writing of EOs 001.01 - 001.04 was complete. Work continues within lesson specifications for EO 001.01 and EO 001.02. Learning assessment plan completed. The Board decided that Enabling Checks (EC) 001.01 and EC001.02 will be home-based assignment checks.

Day #7 - 21 Jul 2015

29. SLt Simmons and PO1 Wood from CFNOS Sub Div joined the Board IOT verify practices in HMC Submarines where Hull Techs are not bourne. They continued to assist in the attachment of appropriate references to teaching points.

30. Completed attaching references to TPs for PO 001.

31. It is noted this QSPWB generated a total of thirteen tasks for approval by NPTG IOT support board objectives, with approval noted in this ROP para 14 on July 15 2015. This will result in a misalignment with the current specialty spec, which included only one vague task. An SPQR will be generated for action.

Day #8 - 22 Jul 2015

32. Mr. Osmond from FMF Cape Scott re-joined the board IOT complete lesson specifications for EOs within PO 002.

33. Mr. Osmond raised the issue about resource allocation, referring to FMF personnel availability for instructing and assessing PO 002. FMF Safety does not necessarily have assisting with this course as part of their business plan and further investigation of this matter is required.

34. Upon further discussion with present board members, the recommendation is made that the QRB composition will be the School's Hull Training Chief (or their delegate), Section Head (acting as board chair), and a qualified Level III Safety and Environment Inspector, as appointed by the FMF Safety Environment Manager. This change ensures a proficient military presence.

35. Chapters 3 & 4 were completed. SMEs were thanked for participation and QSPWB adjourned.

Day #9 - 23 Jul 2015

36. SLt Hynes, SLt Townsend, and SLt Faludi began review of data inserted into ISAT to ensure completion and accuracy. Follow up with PO2 Caldwell will take place as needed.

Day #10 - 24 Jul 2015

37. Post-board review continued.

38. The Hull Tech pre-requisite for attending this course was amended from QL5 to QL4 (their completion of OJPR). The previous QSP required a Hull Tech to be QL5 qualified prior to commencing the Gas Free course; however this was never the practice. The Gas Free course is core training for Hull Techs and is incorporated as a phase of the QL5 course.

28 Sep 2015

39. Further changes were made to the course pre-requisites. The previous version of this QSP allowed DND civilian personnel in possession of a 4th Class Engineering Certificate or above, and personnel employed at an FMF Safety Section to attend this course. In consultation with the CFNES MSE Div Comd, these pre-requisites were removed as it is not the mandate of an NTPG TE to train and grant qualifications to civilians.

Lessons Learned:

1. Nil.

Guidance to Instructional Staff:

1. Nil.

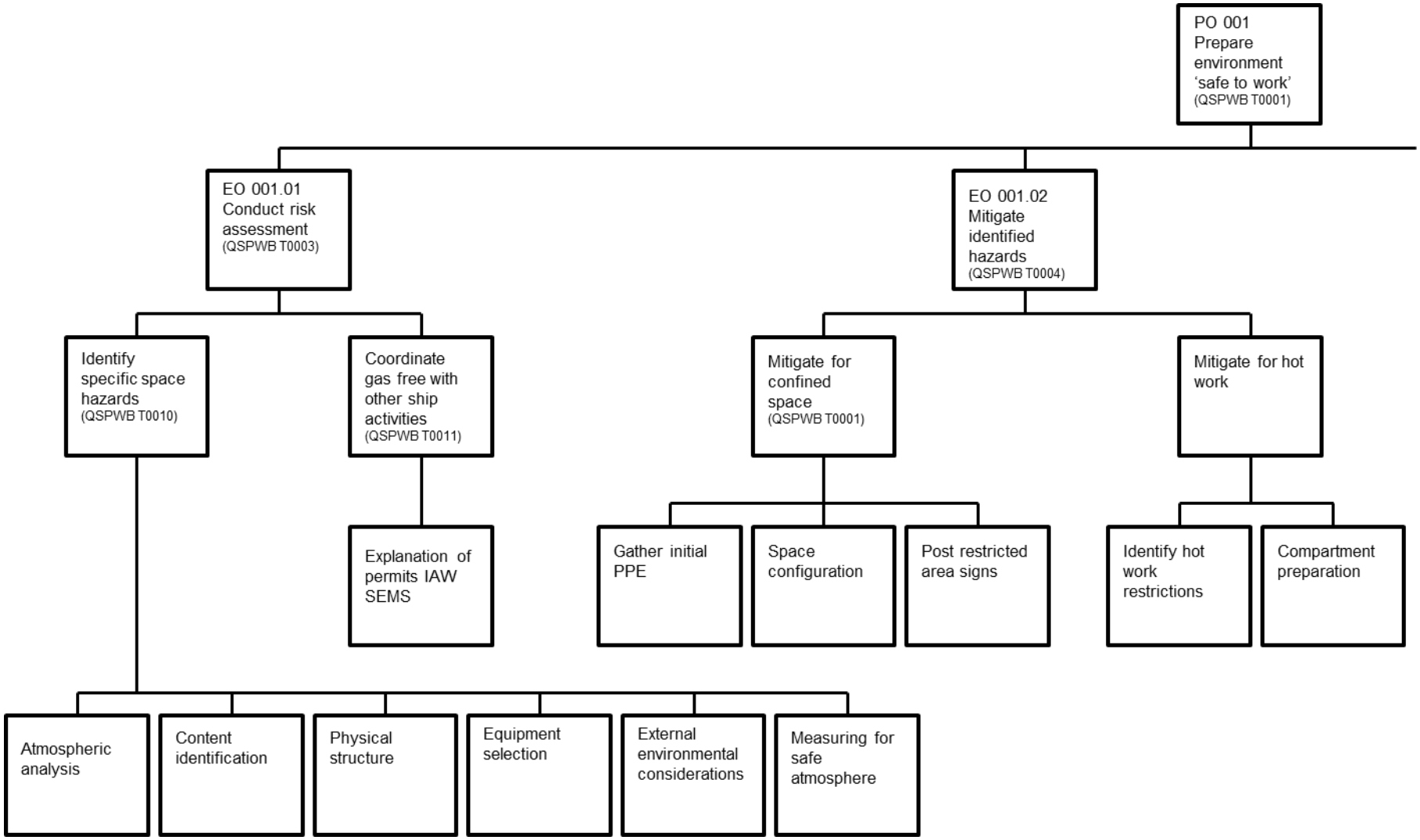
Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Request new MITE Code	SLt Hynes	30 Sept 15
2	Request archive of current MITE Code	SLt Hynes	30 Sept 15
3	Contact Ken Beatty (LCMM) for the necessity to have CO H sensors added for use on submarines	PO2 Caldwell	30 Sept 15
4	LSR submission for DLN Module to include content from PO001	CFNES SO QA	30 Dec 15
5	Discuss with Hull Tech MOC Advisor about	PO2	30 Jul 15

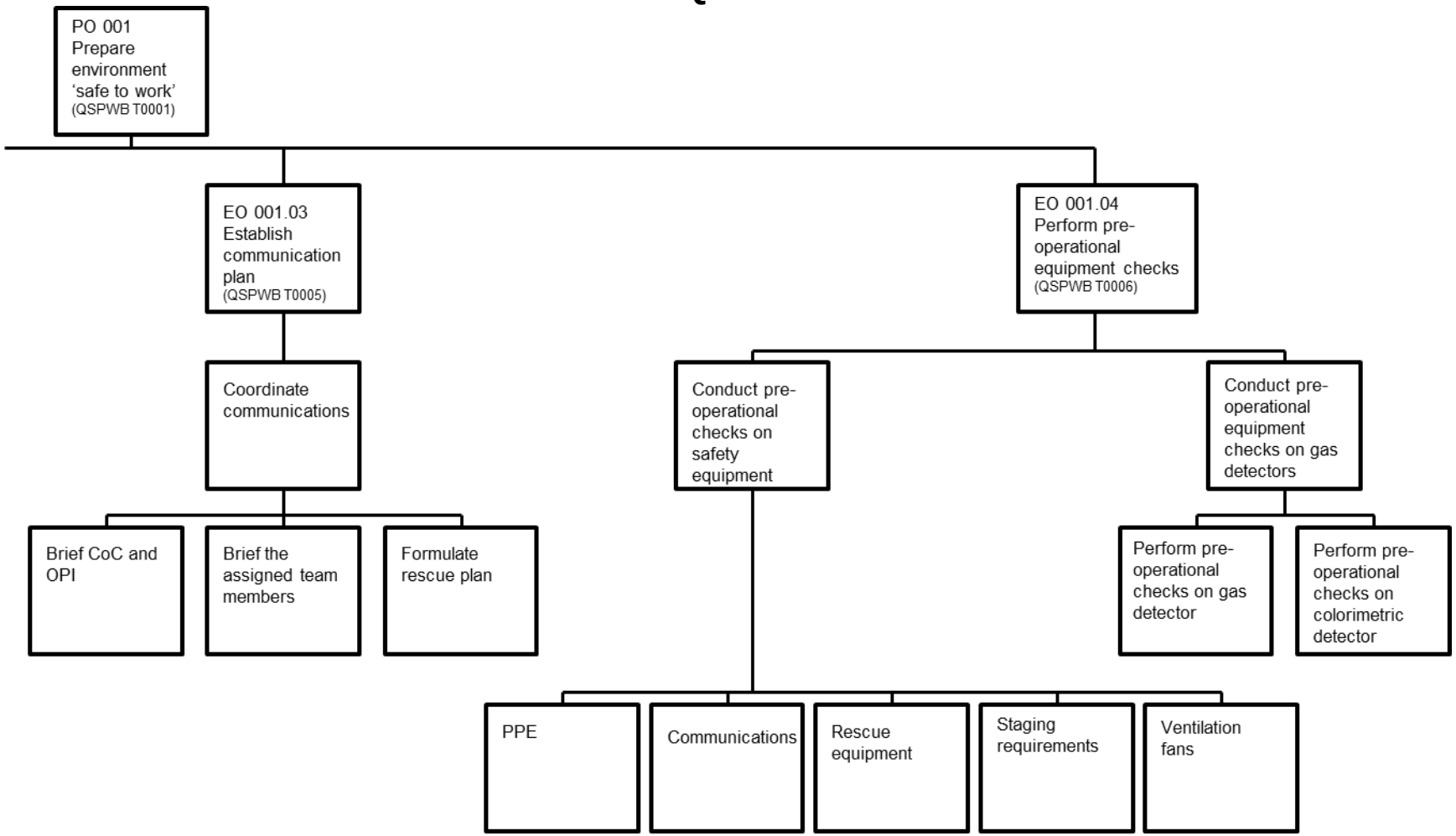
Item	Action	OPI	Expected Date of Completion
	postings to FMF Safety in anticipation of an increased participation in endorsements	Caldwell	
6	SPQR	CPO2 Burton	30 Sept 15

Other Considerations (if applicable): Nil.

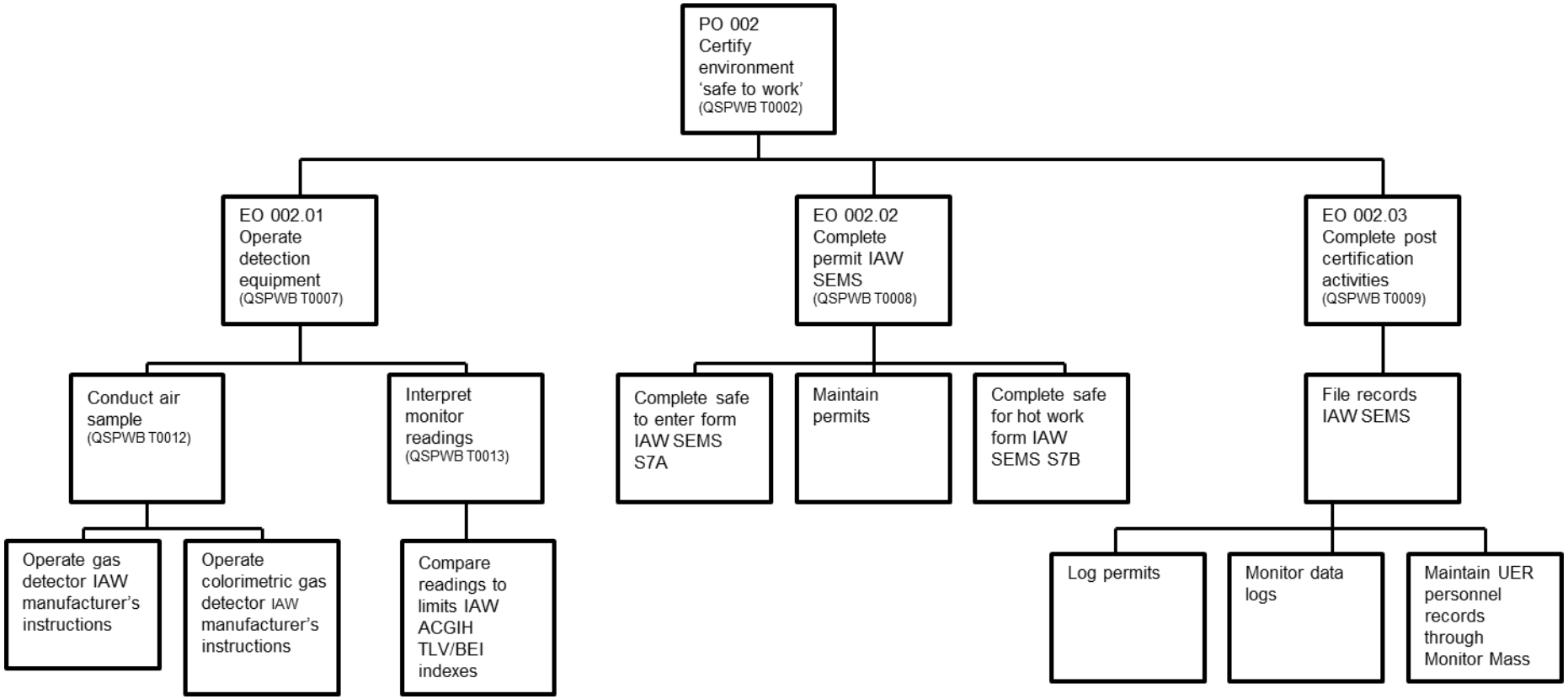
Annex D - QSPWB Scalar



ANNEX D – QSPWB SCALAR



ANNEX D – QSPWB SCALAR



ANNEX E – TRAINING COMPLETION CERTIFICATION

TRAINING COMPLETION CERTIFICATION

I certify that _____ has completed training to the standards stated in this QSP and has passed the Gas Free Certification Qualification Review Board; therefore, is qualified to GAS FREE CERTIFICATION (NQUAL AFPY).

QUALIFICATION REVIEW BOARD MEMBERS

Section Head (Chairperson): _____ _____
Date

School`s Hull Training Chief
(or delegate): _____ _____
Date

Level III Safety and
Environment Inspector
(as appointed by FMF
Safety Environment Manager): _____ _____
Date

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
QSP	Qualification Standard and Plan
Hull Tech	Hull Technician
Mar Eng	Marine Engineer
MESO	Marine Engineering Systems Operator
RCN	Royal Canadian Navy
MITE	Military Individual Training and Education
TA	Training Authority
CCA	Curriculum Control Authority
CFNES	Canadian Forces Naval Engineering School
CFITES	Canadian Forces Individual Training and Education System
EC	Enabling Check
PC	Performance Check
BAT	Base Administrative Time
TTT	Training Travel Time
TAT	Training Administrative Time (TAT)
PT	Physical Training
FMF	Fleet Maintenance Facility
TE	Training Establishment
PLAR	PRIOR LEARNING ASSESSMENT AND RECOGNITION
PO	Performance Objectives
EO	Enabling Objectives
SO	Standing Order
SOP	Standard Operating Procedure
TRB	Training Review Board
PRB	Progress Review Board
DWAN	Defence Wide Area Network
WHMIS	Workplace Hazardous Materials Information System
PCB	Polychlorinated biphenyl
LEL	Lower Exposure Limit
UEL	Upper Exposure Limit
TLV	Threshold Limit value
TWA	Time Weighted Average
STEL	Short Term Exposure Limit
C	Carbon
IDLH	Immediately Dangerous to Life and Health
SEMS	Safety and Environmental Management

	System
PID	Photo Ionization Detector
PPE	Personal protective equipment
CoC	Chain of Command
OPI	Officer of primary Interest
RAT	Rapid attack team
OSC	On Scene Commander
HQ1	Headquarters
ACGIH	American Conference of Governmental Industrial Hygienists
UER	Unit Employment Record
BEI	Biological Exposure Indices

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO / EO	PO / EO Title	EO	EC*	PC*	PO
001	Prepare environment ‘safe to work’	900	25	125	1050
002	Certify environment ‘safe to work’	1050	250	350	1650
	Total (minutes)	1950	275	475	2700

* Includes review and debrief time.

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).

Template Annex I Costing.xls

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	N/A	N/A		N/A	N/A	N/A
R&Q	N/A	N/A	N/A	N/A	N/A	N/A
Incidentals	N/A	N/A	N/A	N/A	N/A	N/A
Subtotal				**$0**	**$0**	**$0.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	1	6.25	3	$0	$6.25	$31.25
Classroom (square meters)	44	$103.8	3	…	…	…
Computer requirements	0	$0	0	$0	$0	$0.00
IT Equip (Proj, Smart board etc)	1	$8000	0	$0	$0	$8000
DWAN connection (Set up)	1	Unknown	0	$0	$0	$0.00
Batteries	1	$12.50	0	$0	$12.50	$62.50
Calibration Gases	1	84.50	0	$0	$84.50	$422.50
Subtotal				**$0**	**$103.25**	**$8516.25**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	0	$.0	0	$0	$0	$0.00
FMF Safety Inspector	1	NIL - MOU	5			
		subtotal		**$0**	**$0**	**$0.00**
	Total Estimated Costs			**$0**	**$103.25**	**$8516.25**

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex B

NPTORD 4500-22
Annex B_QSP Amend

Minutes of Gas Free Certification held at CFNES MSE Div 18 January 2002.

Revision	Approval Date	Authority
1	19 Sept 02	MS: 4500-14 (DMTE 2), 20 Sept 02
QSP-MS: 1:00	1 Feb 06	3371-4975-1 (DMTE 2-5/70341), 10 Apr 06
2:00	17 Nov 09	DMTE 2-5

28 Sept 16 – Approved QSP, SPQR is still outstanding from DGPR as it was submitted by D Nav P&T Feb 2016

MARPAC-#587893-v
1-QSPCR_-_Gas_Free

ANNEX K – ASSESSMENT PLAN

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.01	EO 001.01	To evaluate the trainees knowledge to conduct risk assessment	Take home written exam	80%	PRB process
EC 001.02	EO 001.02	To evaluate the trainees knowledge to mitigate identified hazards	Take home written exam	80%	PRB process
EC 001.03	EO 001.03 – EO 001.04	To evaluate the trainees knowledge to establish a communication plan and perform pre-operational equipment checks	Practical assessment	Pass/Fail IAW EC Checklist	PRB process; Tutorial with instructor prior to PC 001
PC 001	PO 001	To evaluate the trainees knowledge/ability to prepare the environment 'safe to work'	Part A – written exam Part B – practical assessment	Part A – 80% Part B – Pass/Fail IAW Assessment details	Re-test; PRB/TRB process
EC 002.01	EO 002.01 – EO 002.03	To evaluate the trainees ability to certify environment 'safe to work'	Practical assessment	IAW assessment details	PRB process; Tutorial with instructor prior to PC 002
PC 002	PO 002	To evaluate the trainees knowledge to certify environment 'safe to work'	Qualification Review Board	IAW Assessment details	Re-test; PRB/TRB process

ANNEX L – SUPPLEMENTARY DOCUMENTATION

S7A: SAFE TO ENTER/CONDUCT HOT WORK IN A CONFINED SPACE PERMIT

NOTICE: THIS PERMIT IS VALID ONLY FOR THE WORK/TIME DESCRIBED				**PERMIT VALID (DATE and TIME)**			**PERMIT NUMBER**
ON-SITE CO-ORDINATOR;				From			
Supervisor:				To			
CONFINED SPACE ID # :				EMPLOYEE(S) ENTERING		TIME IN	TIME OUT
Confined Space Class	**A**	**B**	**C**				
Confined Spaces location:							
Description of Work:							

TESTING EQUIPMENT

Name: Multi Gas Detector Model: PhD Ultra Ser. No. _____

CALIBRATION

Date: _____ Time: _____ Calibrator: _____

PRE ENTRY TEST RESULTS

TEST	**ACCEPTABLE LEVELS**	**TLV**	**RESULTS**	**SIGNATURE**
1. LEL	50%			
2. Oxygen (O2)	19.5%(Min) 20.9% (ideal) 23% (Max)			
3. Hydrogen Sulfide H2S	2 PPM (Max)	10 PPM		
4. Carbon Monoxide (CO)	5 PPM (Max)	25 PPM		
5. Toxic Gases	None 0%			
6. Others				

NOTE: Maximum LEL for Hot Work is 10% LEL

15 Min Intervals	**LEL**	**O2**	**H2S**	**CO**	**TG**	**Other**

RESCUE EQUIPMENT	YES	NO	TYPE USED	SAFETY REQUIREMENTS	YES	NO
Rescue Procedures				1. Hot Work Permit		
SCBA Breathing system				2. Safety Watcher		
Air Line System				3. Isolate/Post Signs/Barricade Work Site		
Ventilation Alarm Monitor				4. Vehicle Shut off		
Communication				5. Look up confined spaces inventory		
Mechanical Lifting Device				6. Drained/Purged		
First Aid Equipment				7. Ventilated		
Safety line(s)				8. Binding/Blanking		
Personal Alarms				9. Mechanical Lockout/Tagged/Grounding		
Fire Extinguishers				10. Electrical Lockout Tagout		
Life Jackets				11. Fall Arrest Devices		
Explosion Proof Lighting				12. Monitor		
Others:				13. Personnel Protective Equip (PPE)		
				14. Clean shaven (A & B Class)		
				15. Non Sparking Tools		
				16. Ignition Sources Eliminated		
				17. Hazard Assessment		

QUALIFIED PERSON/PERMIT AUTHORIZER	EMPLOYEE SIGNATURE
I certify the completion of all test detailed on this permit, adherence to all confined spaces regulations and that all PPE has been provided and is being utilized. **ON-SITE CO-ORDINATOR'S :** _____ (Print Please) **Date**_____ **Signature**_____ **CHAPTER 2: NOTE** Permit valid for 8 hours only. All copies of permit will remain at job site until job is completed. Return original to the UGSO office following job completion.	I certify that I have read, been briefed on and fully understand the Entry Permit, Hazard Assessment Report, PPE Requirements and Emergency Procedures for entry into this confined space. **Employee (A)**_____ (Print Please) **Signature**_____ **Employee (B)**_____ (Print Please) **Signature**_____ **Employee (C)**_____ (Print Please) **Signature**_____ **Employee (D)**_____ (Print Please) **Signature**_____

ALL PERSONNEL HAVE EXITED THIS CONFINED SPACE AND THAT THE AREA IS SECURED_____
Shop Supervisor SIGNATURE/DATE

ORIGINAL TO UGSO UPON COMPLETION

S7B: SAFE TO CONDUCT HOT WORK PERMIT

CFB HALIFAX "HOT WORK PERMIT"	
DATE:	
BUILDING:	
CONTRACTOR PHONE #:	CONTRACTOR LOCATION:
WORK TO DONE:	
SPECIAL PRECAUTIONS:	
IS FIRE WATCH REQUIRED?	
THE LOCATION WHERE THE WORK IS TO DONE HAS BEEN EXAMINED, NECESSARY PRECAUTIONS HAVE BEEN TAKEN AND PERMISSION IS GRANTED FOR THIS WORK (SEE PRECAUTIONS IN DIRECTIVE #S11)	
PERMIT EXPIRES:	
SIGNED:	
START TIME:	COMPLETED:
FINAL CHECK THE FOLLOWING SHALL BE CHECKED: THE WORK AREA AND ALL ADJACENT AREAS WHERE SPARKS AMD HEAT MAY HAVE SPREAD (INCLUDING FLOORS ABOVE AND BELOW AND ON OPPOSITE SIDES OF THE WALLS).	